Integrala complexă

Fie $A \subset \mathbb{C}$ o mulțime deschisă, $f: A \longrightarrow \mathbb{C}$ o funcție continuă $f = P + iQ$

Fie $\gamma: [a,b] \to A$ un drum (o curbă) de clasă C^1 pe porțiuni

Fie $\gamma(t) = z(t) = x(t) + iy(t)$

$x, y: [a,b] \longrightarrow \mathbb{R}$ funcții de clasă C^1 pe porțiuni

Fie $\Delta_n: a = t_0 < t_1 < t_2 < \dots < t_n = b$ o diviziune a interv. $[a,b]$ ce împarte curba γ în n arce $l_0, l_1, \dots l_{n-1}$

Fie $z_k = \gamma(t_k)$ începutul arcului l_k și punctul $z_{k+1} = \gamma(t_{k+1})$ sfârșitul arcului l_k. Alegem $\xi_k \in [t_k, t_{k+1}]$ arbitrar

Fie $\zeta_k = \gamma(\xi_k) = \xi_k + i\eta_k$

Fie $\sum_{k=0} f(\zeta_k)(z_{k+1} - z_k)$ suma integrală complexă (relativ la $f, \zeta_k, \gamma, \Delta_n$)

Fie $\nu(\Delta_n) := \max\{t_1 - t_0, t_2 - t_1, \dots, t_n - t_{n-1}\}$ norma diviziunii Δ_n

Lemă

$$\lim_{\nu(\Delta_n)\to 0} \sum_{k=0}^{n-1} f(\zeta_k)(z_{k+1}-z_k) = \int_\gamma P\,dx - Q\,dy + i\int_\gamma Q\,dx + P\,dy$$

Demo

$$\sum_{k=0}^{n-1} f(\zeta_k)(z_{k+1}-z_k) =$$

$$= \sum_{k=0}^{n-1} \left(P(\xi_k,\eta_k)+iQ(\xi_k,\eta_k)\right)\left[x_{k+1}-x_k+i(y_{k+1}-y_k)\right]$$

$$= \sum_{k=0}^{n-1} P(\xi_k,\eta_k)(x_{k+1}-x_k) - \sum_{k=0}^{n-1} Q(\xi_k,\eta_k)(y_{k+1}-y_k) +$$

$$+ i\sum_{k=0}^{n-1} Q(\xi_k,\eta_k)(x_{k+1}-x_k) + i\sum_{k=0}^{n-1} P(\xi_k,\eta_k)(y_{k+1}-y_k)$$

f continuă $\Longrightarrow$ P, Q continue $\not\Rightarrow$

$x, y \in C^1$ pe porțiuni

$$\Longrightarrow \sum_{k=0}^{n-1} P(\xi_k,\eta_k)(x_{k+1}-x_k) \longrightarrow \int_\gamma P\,dx$$

$$\sum_{k=0}^{n-1} Q(\xi_k,\eta_k)(y_{k+1}-y_k) \longrightarrow \int_\gamma Q\,dy$$

$$\sum_{k=0}^{n-1} Q(\xi_k,\eta_k)(x_{k+1}-x_k) \longrightarrow \int_\gamma Q\,dx$$

$$\sum_{k=0}^{n-1} P(\xi_k,\eta_k)(y_{k+1}-y_k) \longrightarrow \int_\gamma P\,dy$$

$\Longrightarrow$

$$\Rightarrow \sum_{k=0}^{n-1} f(\zeta_k)(z_{k+1}-z_k) \longrightarrow \int_\gamma P\,dx - Q\,dy + i\int_\gamma Q\,dx + P\,dy$$

Def Integrala complexă a funcției f de-a lungul curbei γ

$$\lim_{\nu(\Delta_n)\to 0} \sum_{k=0}^{n-1} f(\zeta_k)(z_{k+1}-z_k) \overset{not}{=\!=} \int_\gamma f(z)\,dz$$

Corolar $\int_\gamma f(z)\,dz = \int_a^b f(z(t))\cdot z'(t)\,dt \qquad \gamma = \gamma(t),\ t\in[a,b]$

$\parallel$

$z(t) = x(t) + iy(t)$

Demo

$$\int_\gamma f(z)\,dz \overset{\text{lemă}}{=\!=} \int_\gamma P\,dx - Q\,dy + i\int_\gamma Q\,dx + P\,dy =$$

$$= \int_a^b \left[P(x(t),y(t))\cdot x'(t) - Q(x(t),y(t))\cdot y'(t)\right]dt +$$

$$+ i\int_a^b \left[Q(x(t),y(t))\cdot x'(t) + P(x(t),y(t))\cdot y'(t)\right]dt$$

$$= \int_a^b \left[P(x(t),y(t)) + iQ(x(t),y(t))\right]x'(t) + i\left[P(x(t),y(t)) + iQ(x(t),y(t))\right]y'(t)\,dt$$

$$= \int_a^b \underbrace{\left[P(x(t),y(t)) + iQ(x(t),y(t))\right]}_{f(z(t))}\underbrace{\left(x'(t)+iy'(t)\right)}_{z'(t)}dt = \int_a^b f(z(t))\,z'(t)\,dt$$

Prop 1) schimbarea sensului

$$\int_{\gamma^-} f(z)\,dz = -\int_{\gamma} f(z)\,dz$$

γ^- este drumul opus lui γ; $\gamma^-(t) = \gamma(a+b-t)$

2) liniaritatea

Fie $\alpha, \beta \in \mathbb{C}$ și $f, g : A \to \mathbb{C}$ continue

atunci $$\int_{\gamma} (\alpha f(z) + \beta g(z))\,dz = \alpha \int_{\gamma} f(z)\,dz + \beta \int_{\gamma} g(z)\,dz$$

3) aditivitatea

Fie $\gamma_1 : [a,b] \to A$
$\gamma_2 : [b,c] \to A$ drumuri de clasă C^1 pe porțiuni

$\gamma_1(b) = \gamma_2(b)$ și $\gamma = \gamma_1 \cup \gamma_2$

$f : A \to \mathbb{C}$ continuă

atunci $$\int_{\gamma} f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz$$

4) $$\left| \int_{\gamma} f(z)\,dz \right| \leq \int_{\gamma} |f(z)|\,ds$$

5) Limitarea modulului

Fie $M = \sup_{z \in A} |f(z)| < \infty$; L = lungimea drumului γ

atunci $$\left| \int_{\gamma} f(z)\,dz \right| \leq ML$$

ex $\int_\gamma (z-z_0)^n dz \overset{\text{Not}}{=} ?$

$n \in \mathbb{Z}$

$\gamma = \mathcal{C}(z_0, R)$

$\gamma: |z-z_0| = R$

$\gamma: z(t) = z_0 + Re^{it} \quad t \in [0, 2\pi]$ parametrizare curbă

$? \overset{\text{Cond}}{=} \int_0^{2\pi} (Re^{it})^n Rie^{it} dt = R^{n+1} i \int_0^{2\pi} e^{it(n+1)} dt =$

$$= \begin{cases} i\int_0^{2\pi} dt & , n=-1 \\ R^{n+1} \cdot i \cdot \frac{e^{it(n+1)}}{n+1}\Big|_0^{2\pi} & , n \neq -1 \end{cases} = \begin{cases} 2\pi i, & n=-1 \\ 0 & , n \neq -1 \end{cases}$$

Teorema Cauchy

Fie $D \subset \mathbb{C}$ un domeniu simplu conex, $f: D \to \mathbb{C}$ o **funcție olomorfă**, $f = P + iQ$; $P, Q \in C^1(D)$

Fie $\gamma: [a,b] \to D$ un **drum închis**, de clasă C^1 pe porțiuni, fără autointersecții a.î. suportul $K = \overline{\text{Int}}\,\gamma$ **verifică condiția Green-Riemann**

atunci $\int_\gamma f(z)\,dz = 0$

Demo $\int_\gamma f(z)\,dz \overset{\text{lemă}}{=\!=} \int_\gamma P\,dx - Q\,dy + i\int_\gamma Q\,dx + P\,dy \overset{G\text{-}R}{=\!=}$

$$= \iint_K \left(-\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy + i \iint_K \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y}\right) dx\,dy = 0$$

($-\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$, $\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y} = 0$)

f olomorfă $\overset{\text{th. Cauchy-Riemann}}{\Longrightarrow}$ $\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$

$$\frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x}$$

Corolar

Fie $D \subset \mathbb{C}$ domeniu simplu conex

$f: D \longrightarrow \mathbb{C}$ fct. olomorfă

γ_1, γ_2 2 curbe de clasă C^1 pe porțiuni, incluse în D, care au același capăt

atunci $\int_{\gamma_1} f(z)\,dz = \int_{\gamma_2} f(z)\,dz$

Demo $\gamma = \gamma_1 \cup \gamma_2^-$

$$\int_\gamma f(z)\,dz \overset{\text{th. Cauchy}}{=\!=} 0 \quad \Longrightarrow \int_{\gamma_1} f(z)\,dz = \int_{\gamma_2} f(z)\,dz$$

$$\int_\gamma f(z)\,dz = \int_{\gamma_1} f(z)\,dz - \int_{\gamma_2} f(z)\,dz$$

Formula Leibniz-Newton

Fie $D \subset \mathbb{C}$ domeniu simplu conex

$f: D \to \mathbb{C}$ fc. olomorfă

atunci $\exists F$ olomorfă a.î. $F'(z) = f(z)$ și

$$\int_{z_0}^{z_1} f(z)dz = F(z_1) - F(z_0)$$

pe orice drum ce unește punctele z_0 și $z_1 \in D$

ex

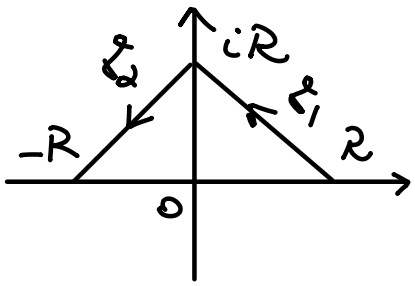
$$\int_{-\bar{u}i}^{\bar{u}i} \cos z \, dz = \sin(\bar{u}i) - \sin(-\bar{u}i)$$

$$= \frac{e^{i\bar{u}i} - e^{-i\bar{u}i}}{2i} - \frac{e^{-i\bar{u}i} - e^{i\bar{u}i}}{2i}$$

$$= \frac{e^{-\bar{u}} - e^{\bar{u}}}{2i} - \frac{e^{\bar{u}} - e^{-\bar{u}}}{2i} = -\frac{1}{i}\operatorname{sh}\bar{u} - \frac{1}{i}\operatorname{sh}\bar{u}$$

$$= 2i \operatorname{sh}\bar{u}$$

ex $J = \int_\gamma \frac{1}{z} dz$ unde $\gamma = \gamma_1 \cup \gamma_2$



$$J = \int_{\gamma_1} \frac{1}{z} dz + \int_{\gamma_2} \frac{1}{z} dz$$

Parametrizarea segmentului ce unește punctele z_1 și z_2

$z(t) = (1-t)z_1 + tz_2 \quad t \in [0,1]$

$\gamma_1: z(t) = (1-t)R + t \cdot iR \quad t \in [0,1]$

$\gamma_2: z(t) = (1-t)iR - t \cdot R \quad t \in [0,1]$

$$I=\int_0^1 \frac{1}{(1-t)R+t\cdot iR}\cdot(-R+iR)\,dt+\int_0^1 \frac{l}{(1-t)iR-tR}(-iR-R)dt$$

$$I=\int_0^1 \frac{-1+i}{(1-t)+it}-\frac{1+i}{(1-t)i-t}\,dt=\int_0^1\left[\frac{-1+i}{(1-t)+it}+i\,\frac{1+i}{(1-t)+it}\right]dt$$

$$I=\int_0^1 \frac{-1+i+i-1}{(1-t)+it}\,dt=2\int_0^1 \frac{(-1-i)(1-t-it)}{(1-t)^2+t^2}\,dt$$

$$I=2\int_0^1 \frac{-1+t+\cancel{it}+i-\cancel{it}+t}{1-2t+t^2+t^2}\,dt$$

$$I=2\int_0^1 \frac{-1+2t}{1-2t+2t^2}+i\,\underbrace{\frac{1}{1-2t+2t^2}}_{2(t^2-t+1/2)=2[(t-1/2)^2+1/4]}\,dt=$$

Notăm $1-2t+2t^2=u$

$(4t-2)\,dt=du$

$$I=\int \frac{1}{u}\,du+\cancel{2}i\int_0^1 \frac{1}{\cancel{2}[(t-1/2)^2+1/4]}\,dt$$

$$I=\ln(1-2t+2t^2)\Big|_0^1+i\cdot\frac{1}{1/2}\operatorname{arctg}\frac{t-1/2}{1/2}\Big|_0^1$$

$$I=\ln 1-\ln 1+2i\operatorname{arctg}(2t-1)\Big|_0^1=2i\left(\operatorname{arctg}1-\operatorname{arctg}(-1)\right)$$

$$I=2i\left(\frac{\pi}{4}+\frac{\pi}{4}\right)=\pi i$$

Formula integrală a lui Cauchy

fie $D \subset \mathbb{C}$ un domeniu

$f: D \longrightarrow \mathbb{C}$ o funcție olomorfă

fie $\overline{\Delta} \subset D$, unde Δ este un domeniu simplu conex mărginit de curba $\gamma = Fr\Delta$ închisă, fără autointersecții, de clasă C^1 pe porțiuni, orientată pozitiv

atunci $\forall\, a \in \Delta$ fixat, avem $\boxed{f(a) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z-a}\, dz}$

Dem funcția $\frac{f(z)}{z-a}$ olomorfă pe $D \setminus \{a\}$ domeniu dublu conex

Alg $\rho > 0$ și $D(a,\rho) \subset \Delta$

Din teorema lui Cauchy pentru domeniu multiplu conex $\Rightarrow$

$$\int_\gamma \frac{f(z)}{z-a}\, dz = \int_{C(a,\rho)} \frac{f(z)}{z-a}\, dz$$

$$\int_{C(a,\rho)} \frac{f(z)-f(a)+f(a)}{z-a}\, dz = \int_{C(a,\rho)} \frac{f(z)-f(a)}{z-a}\, dz + \int_{C(a,\rho)} \frac{f(a)}{z-a}\, dz \quad (\mathrm{I})$$

f olomorfă pe $D \Rightarrow f$ olomorfă în $a \Rightarrow$

$\Rightarrow f$ continuă în a

$\Rightarrow \forall \varepsilon > 0 \ \exists \delta_\varepsilon > 0$ a.î. $\forall z$ cu $|z-a| < \delta_\varepsilon$

avem $|f(z)-f(a)| < \varepsilon$

$$\Rightarrow \left| \int_{\mathcal{C}(a,\rho)} \frac{f(z)-f(a)}{z-a}\,dz \right| \overset{\text{Prop. 4}}{\leq} \int_{\mathcal{C}(a,\rho)} \frac{|f(z)-f(a)|}{|z-a|}\,ds < \frac{\varepsilon}{\rho} \int_{\mathcal{C}(a,\rho)} ds$$

($|z-a| = \rho$)

$$= \frac{\varepsilon}{\rho} \cdot 2\pi\rho \xrightarrow[\varepsilon=0]{} 0$$

$$\Rightarrow \int_{\mathcal{C}(a,\rho)} \frac{f(z)-f(a)}{z-a}\,dz = 0 \overset{\text{I}}{\Longrightarrow} \int_{\mathcal{C}(a,\rho)} \frac{f(z)-f(a)+f(a)}{z-a}\,dz = \int_{\mathcal{C}(a,\rho)} \frac{f(a)}{z-a}\,dz$$

$$\Rightarrow \int_{\mathcal{C}(a,\rho)} \frac{f(z)-f(a)+f(a)}{z-a}\,dz = f(a) \int_{\mathcal{C}(a,\rho)} \frac{1}{z-a}\,dz = 2\pi i f(a)$$

$$\int_\gamma \frac{f(z)}{z-a}\,dz = f(a) \cdot 2\pi i \Rightarrow f(a) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z-a}\,dz$$

Obs

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

$$f'(a)=\lim_{h\to 0}\frac{1}{h}\left[\frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-a-h}dz-\frac{1}{2\pi i}\int_\gamma\frac{f(z)}{z-a}dz\right]$$

$$f'(a)=\lim_{h\to 0}\frac{1}{2\pi i h}\int_\gamma f(z)\left(\frac{1}{z-a-h}-\frac{1}{z-a}\right)dz$$

$$f'(a)=\lim_{h\to 0}\frac{1}{2\pi i h}\int_\gamma\frac{f(z)\cdot h}{(z-a-h)(z-a)}dz$$

$$\Rightarrow f'(a)=\frac{1}{2\pi i}\int_\gamma\frac{f(z)}{(z-a)^2}dz$$

Prin inductie $\longrightarrow$ $$\boxed{f^{(n)}(a)=\frac{n!}{2\pi i}\int_\gamma\frac{f(z)}{(z-a)^{n+1}}dz}$$

ex $\int_{|z-\frac{\pi i}{4}|=\frac{1}{2}} \operatorname{th} z\,dz \overset{\text{Not.}}{=\!=} I$

$$\operatorname{th} z=0\Rightarrow\frac{e^z+e^{-z}}{2}=0\ \Big|\cdot e^z\Rightarrow e^{2z}+1=0\Rightarrow e^{2z}=-1$$

$$\Rightarrow 2z=\operatorname{Ln}(-1)=\{\ln|-1|+i(\arg(-1)+2k\pi),\ k\in\mathbb{Z}\}$$

$$2z=\{i(\pi+2k\pi),\ k\in\mathbb{Z}\}\Rightarrow z=\{i(\tfrac{\pi}{2}+k\pi),\ k\in\mathbb{Z}\}$$

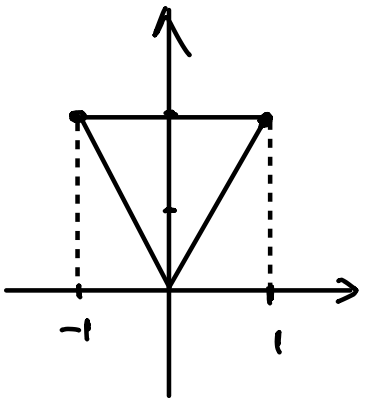
$$\left|z-\frac{\pi i}{4}\right|=\left|i\left(\frac{\pi}{2}+k\pi\right)-\frac{\pi i}{4}\right|=\left|\frac{\pi i}{4}+ik\pi\right|=\left|\frac{\pi}{4}+k\pi\right|>\frac{1}{2},\ k\in\mathbb{Z}$$

$\xrightarrow[\text{Cauchy}]{\text{Th.}}$ $y = 0$

ex $\int_{\gamma} \frac{\operatorname{tg} z}{z-i}\,dz$ $\quad \gamma$ = triunghi cu vârfurile $\{0;\ \pm 1 + 2i\}$

$\cos z = 0 \Rightarrow \frac{e^{iz}+e^{-iz}}{2} = 0 \mid e^{iz} \Rightarrow$

$\Rightarrow e^{2iz} = -1 \Rightarrow$



$\Rightarrow 2iz = \operatorname{Ln}(-1) = \{ i(\pi + 2k\pi),\ k \in \mathbb{Z} \} \Rightarrow$

$\Rightarrow z = \{ \pi/2 + k\pi,\ k \in \mathbb{Z} \}$

$y = 2\pi i f(i) = 2\pi i \operatorname{tg}(i) = 2\pi i \frac{\sin i}{\cos i} = 2\pi i \frac{i \operatorname{sh} 1}{\operatorname{ch} 1} = -2\pi \operatorname{th} 1$

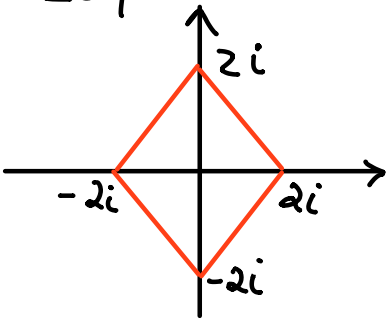
$\sin iz = i \operatorname{sh} z$

$\cos iz = \operatorname{ch} z$

ex $\int_{|z-1|=1} \frac{z^2+1}{z^2-1}\,dz = \int_{|z-1|=1} \frac{\frac{z^2+1}{z+1}}{z-1}\,dz = 2\pi i \cdot f(1) = 2\pi i$ ($f(z)$; a)

ex $y = \int_{\gamma} \frac{z^3 + \sin z}{(z-i)^3}\,dz$ ($f(z)$; a; $u+1$) $\quad \gamma$ pătratul de vârfurile $\{\pm 2;\ \pm 2i\}$

$\Rightarrow y = \frac{2\pi i}{2!} f^{(2)}(i) = \pi i f^{(2)}(i)$

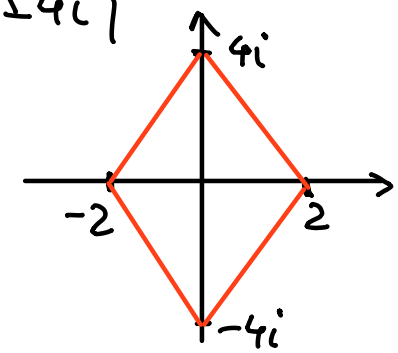


$f(z) = z^3 + \sin z \Rightarrow f'(z) = 3z^2 + \cos z$

$f''(z) = 6z - \sin z$

$y = \pi i\left(6i - \sin i\right) = \pi i\left(6i - i\,\mathrm{sh}\,1\right) = -\pi\left(6 - \mathrm{sh}\,1\right)$

ex $\displaystyle\int_\gamma \frac{\mathrm{ch}\left(z^2 - \pi i\right)}{z - \pi i}\,dz$ & frontiera figurii cu vârfurile $\{\pm 2;\ \pm 4i\}$



$$\int_\gamma \frac{\mathrm{ch}\left(z^2 - \pi i\right)}{z - \pi i}\,dz = 2\pi i f(\pi i)$$

$$= 2\pi i\,\mathrm{ch}\left(-\pi^2 - \pi i\right) = 2\pi i\,\frac{e^{-\pi^2 - \pi i} + e^{\pi^2 + \pi i}}{2} =$$

$$= \pi i\left[e^{-\pi^2}\left(\cos\pi - i\sin\pi\right) + e^{\pi^2}\left(\cos\pi + i\sin\pi\right)\right]$$

$$= -\pi i\left(e^{-\pi^2} + e^{\pi^2}\right) = -2\pi i\,\mathrm{ch}\,\pi^2$$